

School of Computing and Information Systems
The University of Melbourne
COMP90049, Introduction to Machine Learning, Semester 1, 2026

Assignment 2 - Group Contract

Contract due: Monday, May 4 2026, 5 PM

Group Name: Group 36 Team

Group Members:

- Tony Gao – 1761068 – tonygao@student.unimelb.edu.au
- Ying Xu – 1876474 – yxu18949@student.unimelb.edu.au
- Jinyi Gao – 1793033 – jinyigao@student.unimelb.edu.au
- Julius Zhou – 1286438 – zozhou@student.unimelb.edu.au

Research Question:

This project studies short-term readmission risk in the Diabetes 130-US Hospitals dataset, with a focus on identifying patients at high risk of being readmitted within 30 days. Rather than focusing only on overall accuracy, we examine whether early readmission represents a distinct high-risk outcome, whether glycaemic-management signals such as HbA1c testing and diabetes medication changes add predictive value beyond general patient complexity and prior healthcare utilisation, and whether model errors are consistent across clinically relevant subgroups.

RQ1: Do early readmission, late readmission, and no readmission show different clinical, medication, and healthcare-utilisation patterns, and can early readmission be modelled as a distinct high-risk outcome?

RQ2: Do glycaemic-management features, including HbA1c testing and diabetes medication changes, improve early-readmission prediction beyond general patient complexity and prior healthcare-utilisation features?

RQ3: Are model performance and false-negative patterns consistent across clinically relevant subgroups, such as age bands, race, and admission pathway, where sample sizes allow?

Project Overview:

We will use the publicly available Diabetes 130-US Hospitals dataset, which contains diabetes-related inpatient encounters from 130 US hospitals and integrated delivery networks between 1999 and 2008. Each row represents one inpatient encounter rather than a fully independent patient. The readmitted variable will first be analysed in its original three-category form (<30, >30, and NO) during exploratory analysis. The main predictive task will then focus on identifying patients readmitted within 30 days as the high-risk class.

To reduce data leakage, we will use a patient-aware split based on patient_nbr so that encounters from the same patient do not appear in both training and testing sets. We will also check whether encounters with discharge outcomes that make 30-day readmission impossible or clinically non-comparable should be excluded from the main analysis. Preprocessing will include handling missing values, removing identifiers, encoding categorical variables, and grouping diagnosis-related variables into broader categories where appropriate. We will also construct clinically motivated features such as prior healthcare utilisation burden, diagnosis complexity, medication-change indicators, and treatment-complexity measures.

To address RQ2, we will compare several classical machine learning models with one neural network under the same patient-aware validation setup. Planned models include Logistic Regression, Naive Bayes, Decision Tree, Random Forest, Gradient Boosting, and a neural network such as an MLP. Because the <30 class is relatively infrequent, class weighting and decision-threshold adjustment will be explored as modelling strategies rather than treated as separate research questions.

Primary evaluation will focus on recall, precision, F1-score, confusion matrices, and PR-AUC for the <30 class, rather than overall accuracy alone. Subgroup analyses will be reported only for sufficiently represented groups; sparse categories may be combined or omitted to avoid unstable estimates. The discussion will focus on why different feature groups and model families behave differently, what types of errors matter most in a healthcare screening setting, and how false-negative patterns vary across clinically relevant subgroups.

Roles and Responsibilities:

All group members will contribute to interpreting results, reviewing code, checking figures and tables, integrating report sections, and editing the final report. Major methodological decisions will be discussed and agreed upon as a group to ensure consistency across the project.

Member Name	Role and Responsibilities
Tony Gao	Data Preparation and Validation Lead Leads data preparation and validation, including documenting the dataset, defining the cohort and target variable, checking missing values, removing identifiers, implementing the patient-aware train/validation/test split, and coordinating the preprocessing pipeline. This member will also draft the data and preprocessing part of the report.
Ying Xu	EDA and Feature Engineering Lead Leads exploratory data analysis and feature engineering by examining the three readmission outcomes, checking class imbalance and feature distributions, defining feature groups, creating utilisation- and treatment-related features, and preparing core visualisations and summary tables. This member will also draft the dataset description and feature engineering part of the report.

Jinyi Gao	Classical Machine Learning Lead Handles the implementation and tuning of classical machine learning models, including Logistic Regression, Naive Bayes, Decision Tree, and Random Forest or Gradient Boosting, while keeping the evaluation pipeline consistent across models. This member will also prepare the main results table and draft the classical modelling and results part of the report.
Julius Zhou	Neural Network and Error Analysis Lead Leads the neural network experiments, supports class weighting and decision-threshold analysis, and examines subgroup reliability and false-negative patterns. This member will also draft the neural-network-related and error-analysis parts of the report and contribute to the discussion section with the whole group.

Communication Plan:

We will use a shared group chat for quick updates and urgent questions, and a shared document/repository for project materials. Important decisions, task changes, and deadlines must be posted in writing so all members can track them. Members should normally respond within 24 hours on weekdays, or notify the group early if unavailable.

Meeting Schedule:

We plan to hold two regular meetings each week, with each meeting lasting about 45 to 60 minutes. One meeting will focus on progress updates and blockers, and the second will focus on experiments, analysis, and writing. Members are expected to attend on time, prepare updates in advance, and inform the group early if they cannot join.

Decision-Making Process:

We will first discuss decisions as a group and aim for consensus. If views differ, we will compare options against the research questions, workload, and available evidence. If consensus is still not reached, a simple majority decision will be used, and the decision will be recorded in writing.

Work Plan and Timeline:

By May 4, 2026, we will finalise the group contract. By May 8, 2026, we will confirm the dataset, research questions, and evaluation plan. By May 12, 2026, we aim to complete preprocessing and a first baseline model. By May 16, 2026, we aim to complete model comparison and tuning. By May 19, 2026, we aim to finish result analysis, visuals, and the first full report draft. From May 20 to May 21, 2026, we will revise the report, check formatting, and verify code reproducibility before submission on May 22, 2026.

Code of Conduct:

All members will communicate respectfully, complete assigned tasks on time, and be honest about progress and difficulties. Different opinions should be discussed

constructively and supported by evidence where possible. If a member is falling behind, the group should address the issue early, redistribute urgent tasks if needed, and document agreed next steps.

Disagreements or non-responsiveness:

If disagreements or non-responsiveness occur:

1. **Communication Attempt:** Initially, the group will first contact the member directly and clearly describe the issue.
2. **Mediation:** If there is no reasonable response within an agreed time, the matter will be discussed in a group meeting and documented.
3. **Escalation:** If the issue remains unresolved, the group will contact the tutor, and if necessary, the subject coordinator, for formal support.

Signature:

By signing below, each group member acknowledges their commitment to adhere to the terms outlined in this contract.

Student 1: _____ Tony Gao _____ Date: ____04/05/2026_____

Student 2: _____ Ying Xu _____ Date: ____04/05/2026_____

Student 3: _____ Jinyi Gao _____ Date: ____04/05/2026_____

Student 4: _____ Julius Zhou _____ Date: ____04/05/2026_____

This contract is intended to establish clear expectations and promote effective collaboration among group members throughout the duration of the project. Any amendments to this contract should be discussed and agreed upon by all group members.